# Information Letter

**Building Together: A Gamification Framework for Real-Time Casual Social Interaction Among Generation Z in the Mixed Reality Metaverse**


| | |
|---|---|
| **Investigator (s)** | Mr. Kris (Baitong) Xie (Student Investigator), Associate Professor Mohd Fairuz Shiratuddin, Dr Mostafa Hamadi, Dr Joo Yeon Park, and Dr Thao Duong |
| **Contact Person** | Kris (Baitong) Xie (Student Investigator) |
| **Address** | School of Information Technology, Murdoch University, Perth, WA<br>Email: Kris.Xie@murdoch.edu.au<br>Kris.Xie2@murdoch.edu.au |


You are invited to participate in this study.

**Background**

The rapid development of immersive technologies—particularly Mixed Reality (MR), which seamlessly blends physical and virtual environments—has created new opportunities for social interaction within the Metaverse. MR enables real-time, spontaneous, and embodied communication, making it a promising space for meaningful digital engagement.

Meanwhile, gamification—the application of game-like elements in non-game contexts—has proven to be a powerful method for increasing motivation and engagement in education, workplaces, and digital platforms. Prior studies have shown that game mechanics such as avatars, identity expression, achievement systems, and social collaboration can foster immersive and rewarding interactions in virtual spaces.

However, current Metaverse platforms often focus on Virtual Reality (VR) and game-centric models, offering limited support for organic, casual social interaction. There is also limited research on how these interactions unfold in MR environments, especially among Generation Z—digital natives who value self-expression, instant feedback, and immersive connectivity.

Our study aims to address these gaps by exploring how gamification can enhance Generation Z's social interaction in Mixed Reality environments. To do this, we are developing and testing a new design model—the Affordance-Driven Gamification Framework (ADGF)—to support real-time, casual, and inclusive Metaverse experiences.

We invite you to take part in an experimental prototype experience over the next few weeks (dates to be confirmed) to help us evaluate and refine this gamification framework. The project is supervised by Associate Professor Mohd Fairuz Shiratuddin, Dr Mostafa Hamadi, Dr Joo Yeon Park, and Dr Thao Duong, and has been reviewed to ensure it meets all ethical standards.

**Aim of the Study**

This study aims to explore how gamification can enhance casual, real-time social interaction among Generation Z users within Mixed Reality (MR) Metaverse environments. Specifically, it investigates which game mechanics—such as avatar identity, gesture-based collaboration, and immersive feedback—can foster deeper engagement and spontaneous connection. To support this, the study introduces and tests the Affordance-Driven Gamification Framework (ADGF), a design model developed to align game elements with the social and technological affordances of MR platforms. Through participant feedback and interaction with apps, this research seeks to refine the framework and offer practical design guidelines for building socially engaging, inclusive MR Metaverse experiences tailored to the behavioral preferences of Gen Z.

**What Does Your Participation Involve?**

If you choose to participate in this study, you will be invited to take part in a short interactive session using a Meta Quest headset. During this session, you will explore a selected Mixed Reality (MR) application or game designed to encourage spontaneous, casual social interactions in an immersive environment.

As a participant, you will be asked to:

- Complete a brief pilot survey collecting demographic information (e.g., age, gender), your preferred games and social apps, and any conditions that may be negatively affected by immersive visual technologies. This step will be completed online and should take about 5 minutes. It will not take much of your time unless you wish to share additional information about your preferences. In addition, we will send you the links to the apps that will be used in the experiment. If you have sufficient time, you may review them to get some basic background information; however, this is optional.
- Attend an introductory session in a designated room on campus. This session will take approximately 5-15 minutes and will introduce you to the device and the apps used in the study.
- Use the MR app or game for about 25-30 minutes in a supervised, campus-based setting alongside other participants. During the experiment, you will interact with virtual elements and, in some cases, engage with other participants in a multi-player MR environment. There are no strict rules; you are simply encouraged to interact with the MR environment as much as possible-earning points, exploring, and enjoying the experience.
- Participate in a guided group discussion immediately following the session (approximately 20-25 minutes). You will be encouraged to share your experiences of the MR interaction in a safe and respectful environment together with other participants. The investigator will be present to facilitate and guide the discussion.

Inclusion Criteria
To participate in this study, participants must:

- Be between 18 and 28 years old.
- Be familiar with digital or gaming environments, such as Mixed Reality (MR), Virtual Reality (VR), or mobile games.
- Be fluent in English to understand instructions and participate in discussions.
- Be currently located in Perth, Western Australia, and available for in-person participation at Murdoch University.

Exclusion Criteria
Participants will not be eligible if you:

- Have medical or psychological conditions that could be negatively affected by immersive visual technologies (e.g., epilepsy, severe motion sickness, or visual/vestibular disorders).
- Are close friends or peers of the investigator or currently play multiplayer games with other potential participants.
- Are under 18 or over 28 years old.
- Are close friends, peers, or in a personal relationship with the investigator.
- Regularly play multiplayer or cooperative games with other potential participants, as this may influence group dynamics and affect the study’s outcomes.

This study has been approved by the Murdoch University Human Research Ethics Committee (Approval No. 2026/003).

With your consent, screen recordings will be captured during the Mixed Reality (MR) experience for research purposes. These recordings will only capture on-screen activity within the MR environment and will not include participants’ faces, voices, or physical surroundings. All collected data will be securely stored and used solely for the purposes of this study. Your responses and any recorded materials will remain confidential. Only the research team will have access to the identifiable data, and any publications or presentations will use de-identified information so that you cannot be personally identified.

The focus group discussions will be audio recorded to ensure the accuracy of transcription and thematic analysis. Each participant will be assigned a unique pseudonym (e.g., P01–P15) prior to the session, which will be used throughout the discussion and in all subsequent documentation to protect participant identities.

Participants will be reminded at the beginning of each session not to disclose personally identifying information such as names, student numbers, workplaces, or contact details.

Audio recordings will be transcribed verbatim by the research team; no external or third-party transcription services will be used. Any identifiable references that may arise incidentally during the session will be replaced with neutral placeholders (e.g., “[name removed]”) during transcription. Once draft transcripts are prepared, participants will be invited to review and verify the accuracy of their own contributions to ensure their views are correctly represented before analysis begins. Following verification of transcription accuracy, the raw audio files will be permanently deleted within 30 days, leaving only de-identified transcripts for thematic analysis.

Audio recordings and transcripts will be securely stored on encrypted, password-protected University servers and Murdoch University’s OneDrive for Business, accessible only to the Chief Investigator and research team. All data will be managed in accordance with the University’s Research Data Management Policy and retained for a minimum of five years post-publication.

Given the sample size (approximately [32] participants), complete anonymity cannot be fully guaranteed, as voices may be potentially recognisable. To mitigate this, transcripts will omit or generalise any contextual details that could inadvertently identify participants. Findings will be reported in aggregated form, and no direct quotes that could reasonably identify an individual will be used. If excerpts of recordings are included in teaching or presentation materials, voices will be digitally modified (e.g., pitch-shifted) to prevent recognition.

In addition, screen recordings captured during Mixed Reality (MR) gameplay sessions will be reviewed and annotated for behavioural analysis (e.g., gestures, movements, interactions). These recordings will not include facial images or identifiable voices, as they capture only the in-headset MR environment. The purpose of the screen recordings is to complement verbal data from the focus groups by providing objective behavioural context for how participants interact within the MR environment.

All transcripts, annotations, and de-identified data will be stored securely and handled in full compliance with the Australian *National Statement on Ethical Conduct in Human Research* (NHMRC, 2023) and Murdoch University’s Human Research Ethics Procedures.

In accordance with Murdoch University’s ethics guidelines, all research data will be securely stored on a password-protected university server for a minimum of five years post-publication. After this retention period, all identifiable data (such as screen recordings or signed consent forms) will be permanently deleted. De-identified data—meaning all information that could identify participants has been removed—may be retained securely for future research related to immersive technology, gamification, or social interaction design. Any future use of the de-identified data will require approval from a recognised Human Research Ethics Committee and will not include any material that could personally identify participants.

**Voluntary Participation and Withdrawal from the Study**

Your participation in this study is entirely voluntary. You may choose not to take part or withdraw at any time without needing to provide a reason. If you withdraw before or during the MR session or focus group discussion, any data collected from you (including audio and screen recordings) will be deleted immediately and will not be included in the study.

Because group activities involve shared digital interactions, your contributions may appear within shared screen recordings. However, if you choose to withdraw after the session, your MR session data can still be removed from the recordings before analysis. In such cases, your data will be deleted, and no information that could identify you will be retained in any analyses, reports, or publications. Furthermore, during the group discussion session, you retain the right to withdraw any comments you have made, and all related audio data you contributed to the focus group discussion will also be securely deleted.

You may also withdraw your consent for the use of your data within one week after being provided with the opportunity to review it, after which your identifiable data (including voice recordings) will be securely destroyed. Choosing not to participate or to withdraw will have no negative consequences and will not affect any services or support you receive from Murdoch University.

**Your privacy**
Your privacy is very important to us. Your participation in this study and any information will be treated in a confidential manner. Your name and identifying details will not be used in any publication arising out of the research.

**Transcript Checking**
Once the focus group discussions have been transcribed, each participant will be invited to review the transcript of their own contributions. This process, known as transcript checking, allows participants to confirm that their statements have been accurately captured and represented. Each participant will receive a copy of their transcript (identified only by their assigned pseudonym, e.g., P01–P15) via a secure, password-protected file.

Participants will have two weeks to review their transcript and may suggest clarifications, corrections, or request the removal of any statements they do not wish to be included in the final dataset. The research team will incorporate these revisions before data analysis begins. If no response is received within this two-week period, the transcript will be considered accurate and will be included in the analysis.

This procedure ensures that participants have full oversight of their own contributions, that the data accurately reflect their intended meaning, and that all content used in analysis has been voluntarily verified. Transcript checking also supports ethical transparency and participant agency, in line with Murdoch University's Human Research Ethics Procedures and the National Statement on Ethical Conduct in Human Research (NHMRC, 2023).

**Possible Benefits**
While there may be no direct personal benefit to you from participating in this study, your involvement will contribute to important research on how gamification and Mixed Reality (MR) technologies can support meaningful, casual social interactions among Generation Z users in immersive environments.

By participating, you may:

- Gain firsthand experience with emerging MR technologies and social interaction designs.
- Reflect on your own social preferences and engagement styles in digital environments.
- Contribute valuable feedback that will help shape future MR experiences and gamified platforms in the Metaverse.
- Help researchers and designers develop more inclusive, engaging, and user-centered immersive environments for future generations.
- You will also have the opportunity to explore the XR world as a pioneer in this emerging field, experiencing firsthand some of the most advanced XR technologies currently available—including the Meta Quest 3 and Apple Vision Pro - both during and after the experimental session briefly.

Your participation may also support broader advancements in immersive technology, education, and digital social design.

**Possible Risks**
There are no specific risks anticipated with participation in this study. However, if you find that you are becoming distressed or overwhelmed during the Mixed Reality experiences such as due to motion discomfort, sensory overload, or social anxiety, you will be advised to receive support from the university's wellbeing services. Alternatively, we will arrange for you to see a counsellor at no expense to you.

Support is available if you experience any emotional or psychological discomfort:

- **Murdoch University Counselling Service** – Free and confidential support for students.
  - Phone: (08) 9360 1227
  - Email: counselling@murdoch.edu.au
  - Website: www.murdoch.edu.au/counselling

Alternatively, if you prefer to speak with an external service, you may contact:

- **Lifeline** – 13 11 14 (24-hour crisis support and suicide prevention services)
- **Beyond Blue** – 1300 22 4636 (24-hour mental health support line)

You may also request that we connect you with the university's wellbeing team or arrange a counsellor appointment at no expense to you.

**Reimbursement**
To thank you for your time and contribution, you will be entered into a draw to receive one of three $100 gift cards after completing the session. This incentive is offered as a token of appreciation for your participation in the study. The gift cards will be funded from the student investigator's annual research funding provided by Murdoch University, which is an internal source of research support.

**Questions**
If you would like to discuss any aspect of this study, please feel free to contact either Kris Xie via email at Kris.Xie@murdoch.edu.au or Associate Professor Mohd Fairuz Shiratuddin via email at m.fairuz@murdoch.edu.au. Either of us would be happy to discuss any part of the research with you.

Once we have analysed the information from this study, we will email a summary of our findings. You can expect to receive this feedback in 2-3 months.

We would like to thank you in advance for your assistance with this research project. We look forward to hearing from you soon.

If you are happy to participate in the study, please contact Kris via email at **Kris.Xie@murdoch.edu.au** (Kris Xie) to arrange a suitable time for your participation session.

This study has been approved by the Murdoch University Human Research Ethics Committee (Approval xxxx/xxx). If you have any reservation or complaint about the ethical conduct of this research, and wish to talk with an independent person, you may contact Murdoch University's Research Ethics & Integrity on Tel. 08 9360 6677 (+61 8 9360 6677 for overseas studies) or e-mail human.ethics@murdoch.edu.au. Any issues you raise will be treated in confidence and investigated fully, and you will be informed of the outcome.